Avishka Shamendra

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A first year PhD student passionate about security and privacy seeking an opportunity to gain research experience in emerging challenges in the field. Recently, I have become intrigued by the security issues surrounding AI agents, both the security of AI agents themselves and the security risks introduced by their use.

Research Interests:
System Security, AI Security, Web Security, Cloud Security, Trusted Computing, Data Privacy

Education

Stony Brook Univeristy, NY — Aug. 2025 – Present
PhD in Computer Science — **Cumulative GPA: 4.00/4.00**

- CSE 509 System Security - A
- CSE 512 Machine Learning - A
- CSE 519 Fundamentals of Data Science - In progress
- CSE 541 Logic for Computer Science - In progress

University of Moratuwa, Sri Lanka — Nov. 2018 – June 2023
B.Sc. Engineering (Hons.), First Class — **Cumulative GPA: 4.13/4.20**
Computer Science & Engineering — Dean's List: Semester 1, 2, 3, 4, 5, 6, 7 & 8

- **Computer Science & Engineering Gold Medal Winner**: Rank 1/128

Nalanda College, Colombo, Sri Lanka — Graduated in 2017
G.C.E. Advanced Level 2017 (Physical Sciences) — 4As | Z-score: 2.314 (top 0.05%)

- National university entrance examination taken by more than 270,000 students

Research Experience & Publications

TruFaaS - A Trust Verification Framework for FaaS — June 2022 – June 2023
Final Year Research Project, Supervisor - Dr. Sunimal Rathnayake — GitHub

- Proposed a novel architecture for a trust verification framework to calculate and verify the trust of functions on a FaaS platform.
- The framework can be integrated on top of an existing FaaS platform and has the capability to calculate a trust value for each function deployed on the platform.
- During function invocation, before a container is started, a novel trust verification algorithm uses a Merkle tree data structure and the server's Trusted Platform Module (TPM) to verify the trust of the function against the previously calculated and stored trust values.
- Implemented *TruFaaS* on the open-source FaaS platform *Fission* to demonstrate its practical effectiveness.
- Played a critical role in finalizing the overall framework architecture and was primarily responsible for designing the trust verification algorithm.
- Technologies: Go | Docker | Kubernetes | Fission Serverless Framework
- **Publication**: A. Shamendra, B. Peries, G. Seneviratne, and S. Rathnayake, "TruFaaS-Trust Verification Framework for FaaS", in *3rd International Conference on Ubiquitous Security (UbiSec)*, pp. 304–318, 2023, Available: UbiSec 2023 Publication

Work Experience

Senior Software Engineer - WSO2 LLC Feb 2025 – Jul 2025
Full-time Technologies: Java | Go | K8s | Docker | Maven

· Leading the development of the API Governance Component for *WSO2 API Manager*, enabling governance of API behavior and security.

Software Engineer - WSO2 LLC July 2023 – Feb 2025
Full-time Technologies: Java | Go | K8s | Docker | Maven

· Implemented AI API Support to expose services like OpenAI and Mistral as managed APIs in the API Manager.
· Designed and developed the metrics module for *Choreo Connect*, gaining hands-on experience with Kubernetes, Docker, and other cloud-native technologies.

Intern Software Engineer - WSO2 LLC. Dec. 2021 – Aug. 2022
Full-time Technologies: Java | Maven | FHIR

· Designed and developed the *OAuth2/FHIR Scope Management Module* for the *WSO2 Open Healthcare* solution, which helped me grasp security concepts such as OAuth2 and OpenID Connect.

Intern Software Engineer - IXD Labs (Pvt) Ltd. May 2020 – May 2021
Part-time | Startup Technologies: Python | Node | K8s | Docker

· At *IXD Labs* I collaborated with a small team to complete a wide range of projects within a given time frame and gained experience in software development and cloud technologies.

Selected Academic & Personal Projects

UniLabs Inventory Management System July 2021 – Dec. 2021
Academic Project GitHub | YouTube

· Designed and developed a barcode-based automated inventory management system for university laboratories, while gaining exposure to web and mobile development and the complexities of software architecture design.
· Technologies: Django | React | Android | Flutter-BLoC | PostgreSQL

Agro Wholesale Market Dec. 2020 - Feb. 2021
Academic Project GitHub

· Implemented a web-based system which allows farmers to post about their harvested produce for sale and wholesale buyers to privately respond with bids, following MVC architecture.
· Technologies: Node | Express | PostgreSQL | React

Predicting GI Track Anomalies Using KVISAR Dataset July 2020
Academic Project GitHub

· Focused on predicting the presence of gastric intestinal track anomalies such as polyps, z-lines and ulcerative colitis. Used the KVASIR dataset and deep CNN architectures such as InceptionV3, Densenet201 and VGG-19
· Technologies: TensorFlow | Python | Matplotlib

Druggable Protein Prediction April 2023 - May 2023
Academic Project GitHub

· Extracted feature encoding from protein strings to predict whether the protein is druggable.
· Did extensive tests of encoding combinations and ML algorithms to improve the predictability
· Technologies: Sklearn | Python | Matplotlib

COVID-19 Situational Analysis Dashboard March 2021
Personal Project GitHub

· Used to provide COVID-19 case figures to Sri Lankans through the API of the Health Promotion Bureau
· Technologies: React | React-Redux | Google Charts

Mobile Application Projects 2020
Personal Projects GitHub-Timetable App | GitHub-Mora GPA App

· My Timetable App: State of the art application for school children to keep track of their timetable. Had 5000+ downloads on Play Store before the app was retired.
· Mora GPA App: Designed for undergraduates of University of Moratuwa to keep track of their GPA. Had 400+ downloads on Play Store before the app was retired.

Awards & Scholarships

IBM Gold Medal for being the graduand with the highest GPA in the Computer Science & Engineering Honors Degree at the University of Moratuwa, awarded at the General Convocation 2023

IBM Engineering Scholarship for achieving the highest GPA for semesters 3 and 4 in the Faculty of Engineering, University of Moratuwa 2019/2020

Mahapola Scholarship awarded by the Sri Lankan government for being a merit award winner (top 1%) in the GCE Advanced Level examination 2018

Competitions & Hackathons

Top Ten **IEEEXtreme 14.0, 15.0, 16.0 - 2020, 2021, 2022**
Consistently ranked among the top ten in Sri Lanka for three consecutive years

Winners **Decode 1.0 - 2021**
Winners of the programming hackathon organized by the ACM student chapter of UoM

Winners **Dentsu RiseUp Challenge Sri Lanka - 2022**
Pitched *Food Pulse Mobile Application* to minimize food waste in hotels & shops in Sri Lanka by re-distributing excess food

Finalists **TADHack Sri Lanka - 2022**
Reached the finals of TADHack telecom application development hackathon with the *Food Pulse Mobile Application*

Technical Proficiencies

Programming Languages	Java, Python, Go, JavaScript, C#
Cloud-based	Kubernetes, Docker, AWS, Azure
Databases	MySQL, PostgreSQL, Oracle, MongoDB, Cloud Firestore
Web/Mobile Development	Django, Node, Express, React, Flutter
Other	Git, LaTeX, TensorFlow, Maven

Leadership & Volunteering

IT/Marketing Director - IAESTE, Sri Lanka 2022/23
IAESTE is a global non-profit organization offering students international technical internship opportunities. Was responsible for all IT & marketing activities, including the development of IAESTE Sri Lanka website.

Starting Member/ Volunteer Tutor - Danuma Yaathra 2018/19
Danuma Yaathra is a non-profit organization that conducts workshops on STEM subjects in under-resourced schools in Sri Lanka

References available upon request